

Where Did the Agent Go Wrong? Evaluating Failure Attribution Against Counterfactual Rescue

Anonymous ACL submission

Abstract

Self-improving agent loops start by attributing each failure to a step of the episode, usually by asking an LLM to read the trace. We evaluate attribution against counterfactual rescue. Because a failure often has several rescuable steps, we measure for every step how much one correction raises the success of continued replays above uncorrected replays from that step, and score a method by this rescue gain at the step it names. On 300 natural failures from a self-improvement loop in τ^2 -bench retail and airline and in AppWorld, with registered analyses, no LLM-based method beat a zero-cost rule that names the first state-changing action (mean rescue gain 0.42, against 0.20–0.34); asking the judges for the step with the largest rescue gain narrowed the gap without closing it (post hoc). Counterfactual search with 40 replays was no better than one judge call at seven times the cost. The null replays decide the ranking: without them (post hoc), early steps earn credit for chance recoveries and an LLM judge ranks first. Repairs drafted from more accurate steps were practically equivalent on held-out tasks to those drafted from the loop’s own, less accurate analysis; the loop’s proposer read nearly every failing trace either way. Attribution benchmarks need null-controlled ground truth, simple baselines and a test on repairs.

1 Introduction

Agents that improve themselves from their own failures share a first step. An analyst model reads a failed episode, decides where it went wrong, and hands that diagnosis to a proposer that edits the agent’s prompts, tools or code (Shinn et al., 2023; Zhao et al., 2024; Zhang et al., 2026b; Chen et al., 2026; Xia et al., 2026). If the diagnosis points at the wrong step, the proposer may repair the wrong thing. Failure attribution has therefore become a task of its own, with benchmarks of human-labelled multi-agent logs (Zhang et al., 2025; Liu et al.,

2026), taxonomies of failure modes (Cemri et al., 2025, 2026; Zhu et al., 2025), and methods that replay a failure with a step changed (Zhang et al., 2026a; Ma et al., 2025; Lin et al., 2026; Bonagiri et al., 2026; Shah, 2026).

Three questions are rarely asked together. Is an attribution correct in the sense a repair loop needs? Is it worth what it costs? And does a more accurate attribution make the loop’s repairs better? A human label says which step an annotator blames; a repair loop needs a step whose correction would have made the episode succeed. Replay-based methods test that directly but spend agent episodes; an LLM judge spends one call. We evaluate both kinds against a ground truth defined by rescue, report cost next to accuracy, and then hand each attribution to a working loop and measure the repairs it produces.

Defining that ground truth turned out to be the hard part. Our registered plan named the *earliest* step whose single correction beats replays without correction. On a pilot, two independent runs of this protocol agreed on 42% of failures (§3): a failure usually has several steps whose correction rescues it, and which one is found first depends on which corrections an oracle happens to write. We therefore measure a *rescue profile*: for every step, the mean gain in success over null replays across four independent corrections. A method is scored by the rescue gain at the step it names, and a failure is *rescuable* when some step reaches a gain of 0.5. The null replays, which continue the unchanged episode from the same step, turned out to matter most: they succeed often from early steps and rarely from late ones, and without them an LLM judge would rank first among the methods and the rule that wins with them sixth.

We apply the profile to 300 natural failures of agents built by a self-improvement loop, RRSI (Xia et al., 2026), in τ^2 -bench’s retail and airline domains (Barres et al., 2025) and in AppWorld with a code agent (Trivedi et al., 2024), and compare

ten attribution methods: three all-at-once judges of different cost, step-by-step and binary-search judges (Zhang et al., 2025), a root-cause attributor, the trace digester RRSI uses, counterfactual search with up to 40 replays, and two structural rules. Methods, metrics and analyses were registered before any ground truth was computed. A second registered experiment re-runs one round of RRSI at eight of its own states in τ^2 with each attribution, or none, given to the proposer, and deploys every resulting repair on 240 held-out episodes (§6). Our contributions:

- **A ground truth by rescue profile** with null controls. It is stable across independent reruns (of 17 retested failures that both runs found rescuable, 15 had the same decisive step) and, post hoc, across the number of oracle samples and the rescue threshold. Without null replays the conclusion reverses, because uncorrected replays from early steps often succeed by chance (§5).
- **A cost-aware comparison of ten attribution methods** on 300 natural failures. No LLM-based method beat the free first-write rule; the closest, binary search, scored 0.08 lower (exploratory interval $[-0.02, +0.18]$). The registered comparison of counterfactual search with a single judge call found no difference in either τ^2 or AppWorld at seven times the cost, and the cheapest judge matched the most expensive. Post hoc, a judge prompt matched to the metric did not close the gap, and a free rule reading τ^2 's gold actions, as the judges do, beat them by 0.16 to 0.18. Which simple rule wins depends on the domain (§5).
- **A test on repairs.** Repairs drafted from first-write or binary-search steps were practically equivalent on held-out tasks to those drafted from RRSI's own, less accurate analysis ($+0.31$ and $+0.29$ points; 90% intervals within ± 2.5), and counterfactual search's were not better. RRSI's proposers read nearly every failing trace whatever they were shown, so this shows that a step pointer does not help a proposer that reads the traces itself, not that attribution cannot matter to a loop; the largest losses came from single harmful edits, which RRSI's own re-evaluation rejected (§6).

2 Related Work

Failure attribution. Who&When introduced automated attribution of the responsible agent and step in multi-agent logs, with all-at-once, step-by-step and binary-search judges, and found step-level accuracy far below agent-level accuracy (Zhang et al., 2025); Who&When Pro revisits these labels (Liu et al., 2026). Taxonomies organise failures into modes, including adaptive ones (Cemri et al., 2025, 2026), and AgentDebug pairs a taxonomy with targeted feedback (Zhu et al., 2025, 2026); ErrorProbe improves diagnosis from its own history (Li et al., 2026). These evaluations score agreement with human or injected labels; we score the effect of correcting the named step.

Counterfactual attribution. AgenTracer labels failures by replaying corrected steps of injected faults and trains an attributor on them (Zhang et al., 2026a). DoVer and REFLECT validate hypotheses by intervening on a trace (Ma et al., 2025; Lin et al., 2026), CausalFlow and Causal Agent Replay attribute failures by counterfactual replay (Bonagiri et al., 2026; Shah, 2026). Our ground truth uses the same intervention, with two differences that turned out to matter: null replays from the same step, without which a fifth to a quarter of τ^2 replays would count as rescues, and a full profile over steps instead of the first step that flips. The profile is a graded, sampled version of but-for causation (Halpern, 2016; Pearl, 2009).

Self-improving agents. Reflexion, ExpEL and ACE turn failures into lessons or playbooks (Shinn et al., 2023; Zhao et al., 2024; Zhang et al., 2026b); HarnessFix diagnoses and repairs harness flaws (Chen et al., 2026); RRSI digests failed traces, proposes harness edits and keeps them by re-evaluation (Xia et al., 2026). Prediction of failure does not imply that intervening helps (Vasudev et al., 2026). We evaluate the diagnostic step these loops share, on failures one of them produced.

3 Ground Truth: Rescue Profiles

Replays. A failed episode is a sequence of agent steps $0, \dots, n-1$. A *replay from step k* keeps steps $0, \dots, k-1$ and the environment state they produced, and lets the same agent and user simulator continue; a *corrected* replay forces a given action at step k first. The agent runs at temperature 0, but replays still vary, so each outcome is a sample.

Oracle corrections. For each step an oracle (DeepSeek-V4-Pro, thinking off) sees the domain policy, the tools, the episode up to the step and the task’s grading criteria: τ^2 ’s gold actions and required information, or AppWorld’s unit tests with their expected values. It either judges the step correct or writes the action the agent should have taken, using only what the agent knew at that point: the criteria tell it the right outcome, not identifiers or facts the agent had not yet seen. The LLM attributes we evaluate read the same grading report, as RRSI’s analyst does (§4); only counterfactual search works without it.

From earliest step to profile. Our first registered protocol named the earliest step whose correction succeeded on at least 2 of 4 replays more than 4 null replays did. On 20 pilot failures, four independent runs of it named the same decisive step in 42% of pairs of runs (50 of 120), and in 23% of pairs where either run found one. Most disagreements came from the oracle, not from replay noise: a step judged correct in one run and corrected in another, or a correction that worked in one run and a different correction that did not in the next. Before computing any ground truth on the main set, we registered protocol 3 instead.

Protocol 3. At every step k we draw $K = 4$ oracle samples. Each sample that corrects the step gets 2 corrected replays, and the step gets 4 null replays. The rescue gain of step k is

$$R_k = \frac{1}{K} \sum_{i=1}^K c_{k,i} (\hat{p}_{k,i}^{\text{corr}} - \hat{p}_k^{\text{null}}), \quad (1)$$

where $c_{k,i} = 1$ if sample i gives an action the replay can apply, and $c_{k,i} = 0$ if it judges the step correct or its action cannot be applied. A failure is rescuable if $\max_k R_k \geq 0.5$, and then its decisive step is the step with the largest gain (earliest on ties). We also record the earliest step with $R_k \geq 0.5$. Figure 1 shows one profile. On the pilot, step-level gains of two independent runs correlated .74 (retail) and .64 (airline), and the decisive step agreed on 15 of 20 failures; where both runs found one, on 7 of 8, the eighth one step apart.

Scoring a method. A method names one step $\hat{k}$. Its score on a failure is $R(\hat{k})$, the rescue gain at that step, and 0 for an invalid step. The primary metric is the mean of $R(\hat{k})$ over rescuable failures; it rewards naming any step whose correction helps,

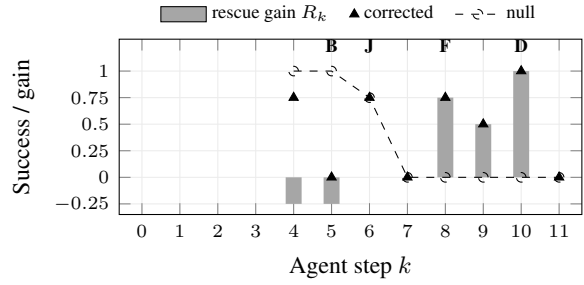


Figure 1: Rescue profile of one retail failure. Corrected: mean success of the oracle’s corrections; null: success of uncorrected replays (run only where some sample proposed a correction). Steps 8–10 all rescue the episode; D marks the decisive step, F the first write, J the all-at-once judges’ step and B binary search’s. At step 6, three of four corrections succeeded, but so did three of four uncorrected replays.

in proportion to how much it helps. Secondary metrics are exact agreement with the decisive step, agreement within one step, agreement with the earliest rescuing step, the share $R(\hat{k}) / \max_k R_k$, and $R(\hat{k})$ over all failures.

4 Setup

Failures. All failures come from RRSI (Xia et al., 2026) run as a self-improvement loop on a DeepSeek-V4-Flash agent: from the harnesses it accepted over 20 rounds, in their evaluations on training tasks and in their deployment on held-out tasks. In τ^2 -bench (Barres et al., 2025) the agent serves a simulated customer through tool calls; in AppWorld (Trivedi et al., 2024) it writes Python against the APIs of nine apps and submits with a final call. We drew 100 failed episodes per domain with a registered rule: no harness error, at most two per task and harness, tasks taken round-robin in a seeded order. They cover 36 (retail), 24 (airline) and 47 (AppWorld) tasks and average 11.7, 10.3 and 19.6 agent steps.

Methods. Every LLM method except counterfactual search reads the failure as RRSI’s analyst reads it: the task (in τ^2 with the simulated customer’s hidden instructions), the episode with numbered steps, and the grading report the agent never sees, which lists τ^2 ’s gold actions and missing information or AppWorld’s failed unit tests with their assertions. The judges therefore have the information the ground-truth oracle has. Each method names one step.

- **Rules:** the *first write*, the first step that calls a

259	state-changing tool or API (else the last step),	and replays. Appendix E lists every change made	308
260	and the <i>last step</i> .	after registration.	309
261	• All-at-once judge (Zhang et al., 2025): one	5 Results	310
262	call over the whole episode, with DeepSeek-	Ground truth. Of 300 failures, 157 were res-	311
263	V4-Flash, V4-Pro, and V4-Pro with thinking.	cuable: 65 of 100 in retail, 37 in airline and 55 in	312
264	• Step-by-step judge: one call per step, in order,	AppWorld. Null replays from the same step suc-	313
265	until a step is judged decisive.	ceeded 27%, 18% and 11% of the time; without	314
266	• Binary search judge: halves the episode	them, many rescues would be chance recoveries.	315
267	$\lceil \log_2 n \rceil$ times, asking which half holds the	Of the rescuable failures, 18, 11 and 26 had two or	316
268	decisive mistake.	more steps with a gain of at least 0.5, so a single	317
269	• Root-cause attributor : one call asking for	"correct step" would have been an arbitrary choice	318
270	a root cause, the earliest wrong step and a	for about a third of them. In the retest, 27 of 30	319
271	component.	failures got the same verdict on rescuability. Of the	320
272	• RRSI trace digester (Xia et al., 2026):	17 that both runs found rescuable, 15 had the same	321
273	RRSI's own multi-turn failure digest; we take	decisive step (the other two, in AppWorld, one and	322
274	the earliest step it cites as evidence.	two steps apart), and step-level gains correlated .80,	323
275	• Counterfactual search (CF search), replay-	.63 and .78.	324
276	based like AgenTracer's labelling (Zhang	No method beat the first-write rule. Table 1	325
277	et al., 2026a) but without grading criteria: one	gives every method's rescue gain. Pooled over	326
278	call names up to five suspect steps with cor-	the three domains, the first write scored 0.42 [0.33,	327
279	rected actions; suspects are tested in order	0.51]; the LLM methods scored between 0.20 (step-	328
280	with 4 corrected and 4 null replays, and the	by-step) and 0.34 (binary search). The paired differ-	329
281	search stops at the first correction that beats	ence between the first write and binary search was	330
282	null by 2 successes or when its budget of 40	+0.08 (exploratory, $[-0.02, +0.18]$), and between	331
283	replays runs out; otherwise it answers with the	the first write and the all-at-once judges +0.11 to	332
284	first suspect. Answers at budgets of 8 to 40	+0.12. Exact agreement with the decisive step or-	333
285	come from the same run.	dered the methods the same way (Appendix Ta-	334
286	Judges are prompted for the decisive mistake, de-	ble 5). Our judges are not unusually weak: on	335
287	defined as the earliest step whose correction would	Who&When they reproduce the published pattern	336
288	most likely have rescued the episode (Appendix B);	(Appendix D).	337
289	a post hoc rerun asks for the step whose correction	Replays did not help. Counterfactual search at	338
290	would most likely rescue it. Pro runs with thinking	budget 40 scored 0.26 pooled. The registered	339
291	off unless stated.	comparison with the all-at-once judge (Pro) found	340
292	Registration and analysis. Methods, prompts,	no difference: $-0.061 [-0.146, +0.032]$ in τ^2	341
293	metrics and analyses were registered, and all meth-	(permutation $p = .20$, $n = 102$) and $+0.025$	342
294	ods had been run, before the main ground truth	$[-0.087, +0.136]$ in AppWorld ($p = .69$, $n = 55$).	343
295	was computed; no method sees ground truth. The	Search cost \$0.022 per failure against \$0.003 for	344
296	registered primary comparison is counterfactual	the judge. It stopped after 17 replays on average,	345
297	search at budget 40 against the all-at-once judge	having confirmed a suspect (42% of failures) or	346
298	with Pro: the paired difference in $R(\hat{k})$ over res-	run out of suspects, and its accuracy stopped ris-	347
299	cuable failures, with a 95% bootstrap interval over	ing at 24 (Appendix Table 6). Its suspects were	348
300	task clusters (10,000 draws) and a paired permuta-	not the problem: on 122 of 157 rescuable failures	349
301	tion test flipping signs by task, on retail and airline	(78%) they included a step with a gain of at least	350
302	pooled and, for AppWorld, separately. Everything	0.5. Its confirmations were. Of the 87 rescuable	351
303	else is estimated with cluster-bootstrap intervals	failures where its own test (4 corrected against 4	352
304	and no further tests; comparisons we did not reg-	null replays, without grading criteria) confirmed a	353
305	ister are marked exploratory. A seeded 10% of	suspect, the confirmed step had a gain of at least	354
306	failures in each domain got a second, independent	0.5 in 41; elsewhere its corrections at the right step	355
307	ground truth. Costs are list-price dollars for tokens	failed, or a weaker step passed the test first.	356

Method	Model	Mean rescue gain $R(\hat{k})$, rescuable failures [95%]				Exact (pooled)	\$ per failure
		Retail (65)	Airline (37)	AppWorld (55)	Pooled (157)		
First write	rule	0.53 [0.38, 0.65]	0.27 [0.13, 0.45]	0.39 [0.26, 0.52]	0.42 [0.33, 0.51]	0.38	0.0000
Last step	rule	0.11 [0.02, 0.24]	0.13 [0.04, 0.23]	0.36 [0.23, 0.50]	0.20 [0.13, 0.29]	0.13	0.0000
All-at-once	Flash	0.29 [0.19, 0.38]	0.41 [0.28, 0.53]	0.25 [0.14, 0.37]	0.30 [0.24, 0.37]	0.28	0.0010
All-at-once	Pro	0.30 [0.19, 0.41]	0.34 [0.21, 0.48]	0.26 [0.15, 0.38]	0.30 [0.23, 0.36]	0.31	0.0031
All-at-once	Pro, thinking	0.26 [0.17, 0.35]	0.42 [0.28, 0.57]	0.28 [0.18, 0.40]	0.31 [0.24, 0.38]	0.32	0.0141
Step-by-step	Pro	0.14 [0.03, 0.25]	0.16 [0.04, 0.25]	0.29 [0.18, 0.40]	0.20 [0.13, 0.26]	0.19	0.0099
Binary search	Pro	0.33 [0.21, 0.47]	0.34 [0.21, 0.46]	0.35 [0.24, 0.45]	0.34 [0.27, 0.41]	0.32	0.0064
Root-cause attributor	Pro	0.26 [0.16, 0.38]	0.36 [0.21, 0.53]	0.22 [0.10, 0.34]	0.27 [0.20, 0.34]	0.28	0.0031
RRSI trace digester	Pro	0.17 [0.07, 0.29]	0.35 [0.24, 0.49]	0.19 [0.11, 0.27]	0.22 [0.16, 0.29]	0.21	0.0077
Counterfactual search@40	Pro, replays	0.22 [0.12, 0.33]	0.31 [0.19, 0.44]	0.28 [0.17, 0.41]	0.26 [0.20, 0.34]	0.26	0.0222

Table 1: Attribution scored by the rescue gain at the named step. Intervals: cluster bootstrap over tasks. Exact: share naming the decisive step. Cost: mean over domains of list-price dollars per failure (tokens and replays).

Cost bought little. Among the all-at-once judges, Flash (\$0.001 per failure), Pro (\$0.003) and Pro with thinking (\$0.014) scored 0.30, 0.30 and 0.31 pooled (Figure 2). Step-by-step judging, at three times the cost of an all-at-once call with Pro, was worse in τ^2 . RRSI’s own digester, which cites evidence steps for a proposer, scored 0.22.

Judges point too early. When a judge missed the decisive step, it usually named an earlier one. In retail, the all-at-once judge with Pro named an earlier step on 40 of 65 rescuable failures and a later one on 7; the step-by-step judge, which stops at the first step it calls decisive, named an earlier one on 49. The first-write rule erred in both directions (11 earlier, 17 later). The registered prompts asked for the earliest decisive mistake, but this explains little: the earliest step with a gain of at least 0.5 was the decisive step in 130 of 157 rescuable failures, and when we re-ran three judges asking for the step whose correction would most likely rescue the task (post hoc, \$3.3), they named steps only slightly later and scored 0.02 to 0.07 higher. The judge with Pro rose from 0.30 to 0.36 (+0.07 [+0.01, +0.12]) and still trailed the first write (−0.06 [−0.17, +0.06]; Appendix Table 7).

The best rule depends on the domain. The first write led in retail (0.53, against 0.14–0.33 for LLM methods) and in AppWorld (0.39), but in airline the all-at-once judges led (0.41–0.42, against 0.27). Appendix Table 4 shows why. In retail, 36 of 65 decisive steps were writes, and the correction replaced a wrong write with the right one; the first write is often that write. In airline, 25 of 37 were utterances, typically a refusal or a clarifying question the agent should have made before acting. In AppWorld,

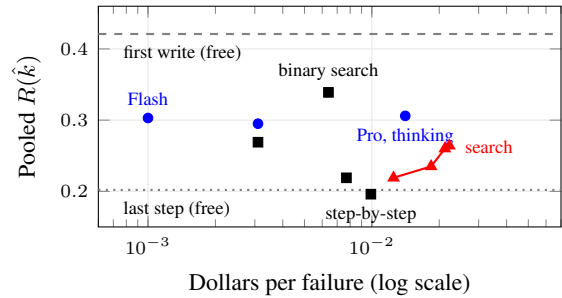


Figure 2: Accuracy against cost. Circles: all-at-once judges; squares: other LLM methods; triangles: counterfactual search at budgets 8, 16, 24 and 40. The free first-write rule lies above every method.

29 of 55 were writes but only 16 were the first one, and 9 were the final submission, made with a wrong answer or before the work was done; decisive steps lay on average 77% of the way through the episode, so naming the last step scored 0.36, against 0.11–0.13 in τ^2 . Binary search was the only LLM method whose accuracy held across domains (0.33, 0.34, 0.35).

A free rule that reads the gold actions. The judges read τ^2 ’s gold actions, and a free rule can too: name the first state-changing call that matches no unused gold action, else the last step (post hoc; τ^2 only). On the 102 rescuable τ^2 failures it scored 0.50 [0.38, 0.60], above the first write (+0.06 [+0.01, +0.12]) and the judges with Pro (+0.16 to +0.18; Appendix Table 8). The oracle corrects from the same gold actions, so part of this lead may be shared information rather than diagnosis.

The ranking is robust, except to the null control. The ground truth rests on three choices: four or-

Ground truth	Resc.	First wr.	Best LLM	Rank	τ
Registered	157	0.42	Binary search 0.34	1	1.00
$K = 2$	188	0.37	Binary search 0.30	1	0.89
$K = 1$	181	0.39	Binary search 0.33	1	0.82
$\theta = 0.25$	226	0.31	Binary search 0.26	1	0.91
$\theta = 0.375$	177	0.38	Binary search 0.31	1	0.96
$\theta = 0.625$	116	0.50	Binary search 0.38	1	0.78
$\theta = 0.75$	107	0.54	Binary search 0.39	1	0.82
One rescuing step	102	0.33	Binary search 0.27	1	1.00
No null replays	203	0.37	Judge, thinking 0.42	6	0.60

Table 2: Method scores under other ground-truth choices (post hoc), pooled over the three domains: rescuable failures, the first write’s mean $R(\hat{k})$, the best LLM method and its $R(\hat{k})$ (judge, thinking: all-at-once with Pro and thinking), the first write’s rank among the ten methods of Table 1, and Kendall’s τ of the method ordering against the registered one ($K = 4$ oracle samples, null replays, threshold $\theta = 0.5$). Rows with fewer samples average over all subsets of the four.

acle samples per step, a rescue threshold of 0.5, and null replays. Table 2 re-scores every method under other choices (post hoc). Using two or one of the four samples (averaged over all subsets), thresholds from 0.25 to 0.75, or only the 102 failures with a single rescuing step left the first write first and binary search the best LLM method, with Kendall’s τ of 0.78 to 1.00 between the method ordering and the registered one. Dropping the null replays did not. Scoring a step by the success of its corrected replays alone made 203 failures rescuable instead of 157, moved the first write from first to sixth place, and put the all-at-once judge with thinking first (0.42 against 0.37). The reversal is partly arithmetic. Without nulls a step scores $R_k + (m_k/K) \hat{p}_k^{\text{null}}$, where m_k of the K samples corrected it; on the 157 registered rescuable failures the added term averaged 0.08 to 0.13 for the LLM methods and 0.04 for the first write. In retail the first write’s lead over the best LLM method shrank from 0.20 to 0.02, and in airline the judge with thinking would lead it by 0.31. The cause is visible in Figure 3: uncorrected replays from the first fifth of an episode succeeded 41–44% of the time, and from the last fifth 2–6%. LLM methods other than binary search named steps on average 53–67% of the way through an episode, against 77% for decisive steps, so without the null control they collect credit for recoveries that happen anyway.

6 Do Better Attributions Give Better Repairs?

Design. Accuracy against rescue profiles is a proxy: a loop needs attributions only to repair its

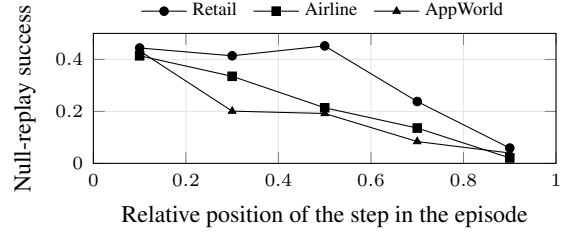


Figure 3: Success of uncorrected (null) replays from a step, by the step’s position in the failed episode (fifths), at the steps where the oracle proposed a correction. Each point averages 27 to 35 steps in the first fifth and 81 to 323 in the others.

agent. We tested this on RRSI itself, in a separately registered experiment. We took two other runs of the loop in each τ^2 domain and the harness each run had accepted by the start of rounds 5 and 15, giving eight incumbent states. At each state we re-ran one unmodified RRSI round five times, changing only the evidence the proposer gets:

- **No attribution:** a note asking the proposer to read the traces itself.
- **RRSI analysis:** the round’s own analysis report and trace digests, unchanged.
- **First write, binary search (Pro) and counterfactual search@40:** for each failing trace, the step the method named and an excerpt of it, with no reason, component or method name (Appendix B).

The proposer works from the traces RRSI builds from the incumbent’s evaluation on training tasks (81 failing traces in all, 42 retail and 39 airline) and, as in RRSI, can read each of them, grading report included. Each round drafts two candidates and puts them through RRSI’s critic, smoke test and full evaluation on training tasks, after which RRSI’s own rule decides whether to accept one. A candidate is *deployable* if it passes these gates with a harness error rate of at most 2%. Every deployable candidate and every incumbent was deployed on 240 held-out episodes (retail 40 tasks $\times$ 6, airline 20 $\times$ 12).

Analysis. A candidate slot’s gain G is its held-out success minus its incumbent’s, in points, and 0 if it was not deployable, since the loop then keeps the incumbent. A contrast between two groups is the mean over the eight states of the difference in their mean gains. The six primary contrasts compare each step attribution with no attribution and with

Evidence for the proposer	$R(\hat{k})$ (Tab. 1)		Mean G , all 16 slots			Deployable		RRSI	Round
	retail / airline	Deployable	both	retail	airline	mean G	$G > 0$	accepted	level
No attribution	–	10/16	–4.38	–9.27	+0.52	–7.00	50%	2	–0.10
RRSI analysis	0.17 / 0.35	13/16	–0.34	–0.31	–0.36	–0.42	38%	1	–0.10
First write	0.53 / 0.27	10/16	–0.03	+0.21	–0.26	–0.04	60%	2	+0.47
Binary search	0.33 / 0.34	9/16	–0.05	+0.16	–0.26	–0.09	44%	0	0.00
Counterfactual search@40	0.22 / 0.31	12/16	–2.47	–0.26	–4.69	–3.30	25%	2	+0.16

Primary contrast	Difference [95%]	p	Holm	Equivalent	Round level [95%]
First write – none	+4.35 [–5.97, +14.67]	0.52	1.00	no	+0.57 [–0.19, +1.34]
Binary search – none	+4.32 [–6.90, +15.55]	0.66	1.00	no	+0.10 [–0.14, +0.35]
CF search@40 – none	+1.90 [–10.24, +14.04]	0.91	1.00	no	+0.26 [–0.99, +1.51]
First write – RRSI	+0.31 [–0.56, +1.18]	0.65	1.00	yes	+0.57 [–0.32, +1.46]
Binary search – RRSI	+0.29 [–0.82, +1.39]	0.60	1.00	yes	+0.10 [–0.14, +0.35]
CF search@40 – RRSI	–2.14 [–5.67, +1.40]	0.31	1.00	no	+0.26 [–0.99, +1.51]

Table 3: Repairs drafted from each kind of evidence in τ^2 : 8 states $\times$ 2 candidates per group. G : held-out success of a candidate minus that of its incumbent, in points, 0 if the candidate was not deployable. $R(\hat{k})$: the method’s rescue gain on rescuable failures in Table 1 (for RRSI’s analysis, its digester). Accepted: candidates RRSI’s own rule accepted. Round level: gain of the accepted candidate, 0 if none. Contrasts: mean over states of group differences, p from within-state permutation; Equivalent: 90% interval within ± 2.5 points. Round-level contrasts are tested by exact sign flips (text).

RRSI’s analysis; p values come from permuting group labels within states and are Holm-adjusted, and 95% intervals from a t distribution over states (df 7). A contrast whose 90% interval lies within ± 2.5 points, about one standard deviation of a single candidate’s gain in earlier runs, counts as practically equivalent. The design, with an expected standard error of 0.9 points per contrast, was registered before any attribution was run on these traces, and no held-out result was looked at until all 62 deployments had finished.

No contrast was significant. Of 80 drafts, 54 were deployable, 9 to 13 per group (Table 3); the others made no proposal, were rejected by the critic or failed the smoke test, and none failed the error check. No primary contrast was significant (Holm-adjusted $p = 1.0$ for all six). Repairs drafted from first-write and binary-search steps were practically equivalent to those drafted from RRSI’s analysis: +0.31 [–0.56, +1.18] and +0.29 [–0.82, +1.39] points. Counterfactual search was 2.14 points below RRSI’s analysis [–5.67, +1.40], mostly through two airline candidates that lost 23 and 14 points, a write gate and a rewrite of baggage updates. The contrasts with no attribution span about ± 10 points because of a single candidate drafted without attribution. Its write gate looked for earlier order lookups in a form they were never recorded in, so it blocked nearly every write without raising an error and lost 72.5 points. Without

it, the no-attribution group’s mean gain was +0.17 points, in line with the others (exploratory).

Accuracy did not carry over. In retail, the first write had located decisive steps three times as well as RRSI’s digester (0.53 against 0.17), yet repairs drafted from the two differed by +0.52 [–0.49, +1.54] points. In airline the digester was as accurate as binary search and more accurate than the first write (0.35, 0.34, 0.27), so there the groups differed in evidence more than in accuracy. The three step methods rarely named the same step (all three agreed on 10% of retail traces and on none in airline), so the proposers did receive different evidence. Most repairs did not help on held-out tasks in any group: 25–60% of deployable candidates gained, and no group’s mean gain was above zero. RRSI’s own rule rejected all three candidates that lost more than 10 points, so at the level of the loop the gain of the accepted candidate differed little between groups: between –0.10 and +0.47 points, with no contrast significant (exact sign-flip $p \geq .25$).

Proposers read the same traces. The proposers’ action logs suggest why the evidence mattered so little (post hoc). In every group the proposer opened 97–100% of the failing traces, with 13 to 20 trace reads per draft, and most reads returned the whole trace or a long window of it. Without any attribution, 96–97% of the reads of a failing

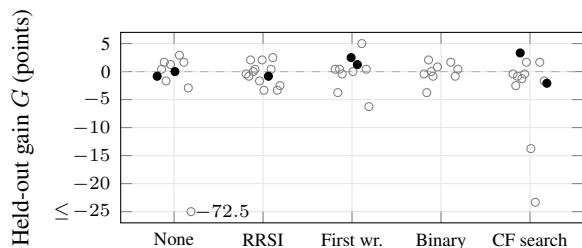


Figure 4: Held-out gain of every deployable candidate in the repair experiment, by the evidence its proposer got. Filled: accepted by RRSI’s own rule. One candidate (−72.5) is drawn at −25.

trace covered the step each of the three methods named for it. Step digests changed how proposers read, raising the share of reads limited to at most five steps from 7–8% to 16–23%, but not what they saw. The experiment therefore tests a step pointer given to a proposer that reads the traces itself, not a loop whose proposer cannot. Figure 4 shows where the differences between groups came from: most candidates in every group landed within 5 points of their incumbent, and the group means were set by a few large losses.

7 Discussion

Replay labels need null controls. Counterfactual replay is a common way to label or verify attributions (Zhang et al., 2026a; Ma et al., 2025; Bonagiri et al., 2026; Shah, 2026). Without a control it is biased: from the first fifth of a failed τ^2 or AppWorld episode, an unchanged continuation succeeded in about two cases of five, so a corrected replay’s success there is weak evidence that the correction mattered. A protocol that counts such successes as rescues favours early steps and the attributors that name them, and in our data it would have reversed the comparison between LLM judges and the first-write rule. Null replays from the same step cost a few replays per step and remove this bias.

Why a rule wins. The rescue profile asks where a single correction helps most. In customer service, an agent that changes the database wrongly fails at that write, and correcting it is the most direct rescue; the first write is often that step. The judges, reading the same gold actions, tended to blame earlier steps: imperfect questions or confirmations that a correction did not turn into success. Where the decisive mistake is a conversational act, as in airline, the judges have the advantage. A judge

that loses to a free rule in one domain and beats it in another needs a domain baseline before its accuracy means anything.

What the oracle’s knowledge does. Our corrections come from an oracle that sees grading criteria. This makes rescue achievable, but it also favours steps where knowing the expected outcome suffices to correct the action, such as writes with the right arguments. The judges see the same criteria, so the gap to the first-write rule is not an information gap. Counterfactual search, which corrects without them, usually had a rescuing step among its suspects but could not confirm it with its own corrections; an attributor that replays needs corrections as good as the oracle’s before replays pay for themselves.

Implications for self-improving loops. The analyst in RRSI-style loops is a trace digester; on our failures it was among the least accurate attributors, yet replacing its analysis with a more accurate step did not give better repairs (§6). A proposer that reads every failing trace in full sees the decisive steps whether or not it is told where they are; attribution may matter more where a loop cannot show its proposer whole traces, with long episodes or many failures per round. And a repair’s value depended less on where the failure lay than on how the edit was built: the largest losses came from write gates and argument rewrites that misfired broadly, and RRSI’s re-evaluation rejected each of them. In a loop like this one, the evaluation that decides which repairs to keep protects the agent more than the accuracy of the attribution behind them.

8 Conclusion

Against a ground truth defined by counterfactual rescue with null controls, LLM attribution of agent failures did not beat a free structural rule on 300 natural failures in three domains; replays, larger models and a prompt matched to the metric did not close the gap. More accurate steps did not make a self-improvement loop’s repairs measurably better, in a loop whose proposer read nearly every failing trace anyway. A rescue profile is stable enough to compare attribution methods, but only with null replays. We release the failures, rescue profiles and code.

Limitations

All agents, oracles and attributors use one model family (DeepSeek-V4). The failures come from one self-improvement loop and three domains, with 37 to 65 rescuable failures each, so per-domain intervals are wide. Ground-truth corrections are written by an oracle that sees grading criteria, which may favour write actions (§7); a different oracle would give a different profile, and the retest measures only the oracle’s sampling variation, on 17 failures that both runs found rescuable (2 in air-line). The threshold of 0.5 for rescuability and the use of four samples per step were fixed on a 20-failure pilot; the ranking held for thresholds from 0.25 to 0.75 and for one or two samples, but these checks and the analysis of the proposers’ reading are post hoc. Our registered prompts asked for the earliest decisive mistake while the primary metric rewards the largest gain; a post hoc rerun with the matched definition raised the judges’ scores by 0.02 to 0.07, so the registered comparison understates them slightly. Fourteen retail failures were drawn after the others, from the two further runs of the loop that the repair experiment also used (one trace is in both), because the first run did not yield enough distinct retail failures. Two of 6,540 AppWorld replays could not run (Appendix E). Who&When is a multi-agent benchmark with a different format; only methods without replay were run there. The repair experiment covers τ^2 only, with eight states and 16 candidate slots per group. For the first write and binary search it rules out differences from RRSI’s analysis larger than 2.5 points, not smaller ones, and its contrasts with no attribution are dominated by one candidate. Its digests carried only the step, not a reason or a component, and its proposers could read every trace with its grading report, so it tests whether pointing a proposer at a better step helps, not whether a better diagnosis would.

Ethics Statement

All environments use synthetic users and data. Attribution methods that misdiagnose failures can lead self-improving agents to change the wrong behaviour; our results suggest that such diagnoses should be checked against simple baselines before they drive automatic changes.

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A Ground-Truth Details

Replays. τ^2 replays restore the database state after the kept steps and continue with the same agent harness and user simulator (DeepSeek-V4-Flash, temperature 0); a corrected action replaces the agent’s turn at step k (a message, or tool calls whose results are then computed). AppWorld replays restore the app state by re-executing the kept code and force the corrected turn, which must contain one Python block; the episode continues with the incumbent harness. A correction the replay cannot apply counts as giving no gain.

Retest. The retest failures were drawn with a seeded random choice of 10% per domain before any ground truth was computed and run with fresh oracle samples and replays.

Cost. The ground truth cost \$11.1 for the first 186 τ^2 failures and \$13.7 for AppWorld (plus \$0.7 for its retest); all parts of the 14 τ^2 failures added later cost about \$1.5. The attribution study, including methods, search, pilots and Who&When, cost about \$58, and the post hoc rerun of three judges with the largest-gain definition \$3.3. The repair experiment (§6) cost about \$35: \$6.4 for drafting and critique, \$1.8 for attribution and search replays, \$7.9 for RRSI’s evaluations and smoke tests (5,639 episodes) and \$18.9 for held-out deployment (14,880 episodes).

B Prompts

The τ^2 prompts follow, verbatim; braces mark what is filled in. The AppWorld prompts replace the description of the agent and the task, the grading report and the component descriptions with AppWorld’s (the code agent, its unit tests, and the same nine components worded for it), and the oracle and search ask for one assistant turn with one Python block.

Shared introduction (all judges)

A customer-service agent (an LLM inside a harness) failed a task of tau2-bench. Below is the failure as the developers’ analysis tools see it: the customer’s hidden instructions, the conversation (each agent step is numbered), and the grading.

The DECISIVE MISTAKE is the earliest agent step such that, had the agent acted correctly at that step and continued normally, the task would most likely have succeeded.

Harness components (a fix to the harness changes one of these):

- prompt: the system prompt text (instructions and policy
→ wording the model reads)
- control_flow: the agent loop code: how the next action is
→ produced, checks or gates applied to tool calls before they
→ run, retries, enforced confirmations
- config: numeric settings such as temperature, token limits or
→ step limits
- output_plumbing: how tool results and tool descriptions are
→ rendered to the model
- context_mgmt: which parts of the conversation history the
→ model sees
- client_tool: extra tools the harness gives the model (e.g. a
→ checker or calculator)
- skill: instruction files loaded for specific situations
- memory: structured state the harness keeps across steps
- subagent: an extra model call, e.g. a second model reviewing
→ an action

```

=== FAILURE ===
{the failure: hidden instructions, numbered steps, grading
→ report}

```

All-at-once

Find the decisive mistake and the harness component whose
→ change would most likely have prevented it. Answer with
→ JSON:

```

{"step": <step number>, "top3": [<the three most likely step
→ numbers, best first>],
"component": "<component>", "components3": [<three most
→ likely components, best first>],
"reason": "<one sentence>"}

```

Step-by-step (asked for $k = 0, 1, \dots$ on the episode up to step k)

Judge step $\{k\}$ only, given what happened before it. Is the
→ agent's action at step $\{k\}$ the decisive mistake (the
→ earliest step whose correction would most likely have made
→ the task succeed)? Answer with JSON:

```

{"decisive": true | false, "component": "<component, if
→ decisive>", "reason": "<one sentence>"}

```

Binary search (episode up to step hi)

The decisive mistake is assumed to lie between step $\{lo\}$ and
→ step $\{hi\}$. Is it in the first half (steps $\{lo\}$ to $\{mid\}$) or
→ in the second half (steps $\{mid + 1\}$ to $\{hi\}$)? Answer with
→ JSON:

```

{"half": "first" | "second", "component": "<the component most
→ likely at fault>",
"reason": "<one sentence>"}

```

Root-cause attributor

An LLM agent failed a task. Find the root cause of the failure.

```

Agent components (id: role):
{COMP_TEXT}

```

```

Trace:
{view(rec)}

```

Give (1) a one-sentence root cause describing the cause, not
→ the symptom;
(2) the index of the earliest step where the agent went wrong;
(3) the single component whose change would most plausibly
→ prevent this failure.
Answer with JSON: {"root_cause": "...", "step": <int>,
→ "component": "<component id>"}

Ground-truth oracle (system prompt)

You construct ground truth for failure attribution. A
→ customer-service agent failed a task of tau2-bench. You see
→ the domain policy, the agent's tools, the task's grading
→ criteria (the database must end as if exactly the gold
→ actions were executed, and the agent must tell the user
→ every required piece of information), and the conversation
→ up to one step of the agent.

Decide whether the agent's action at that step is a mistake: an
→ action a careful agent following the policy would not take
→ on the way to the gold outcome. Actions that are correct,
→ or harmless and do not lead away from the gold outcome, are
→ ok.

If it is a mistake, give the single action a careful agent would
→ take at that step instead: one assistant turn, either a
→ message to the user or one or more tool calls. Correct only
→ this step; do not do more than a careful agent would do at
→ this point. The corrected action must be one the agent
→ could have taken with what it knew then (the conversation
→ and tool results so far, and the policy); use the grading
→ criteria only to know what the right outcome is, never to
→ supply ids, amounts or facts the agent had not yet learned.

Answer with JSON:

```

{"verdict": "ok" | "mistake", "why": "<one sentence>",
"action": {"message": "<text to the user>" | {"tool_calls":
→ [{"name": "<tool>", "arguments": {...}]}}}
("action" only when the verdict is mistake.)

```

Counterfactual search, suspect call (system prompt)

A customer-service agent on tau2-bench failed its task: the
→ customer's request was not resolved as the domain policy
→ requires. You see the domain policy, the agent's tools and
→ the conversation; you do not know what the correct outcome
→ was.

Name the steps where the agent most likely went wrong, most
→ suspect first (at most 5). For each, give the single action
→ a careful agent would have taken at that step instead (one
→ assistant turn: a message to the user or one or more tool
→ calls, using only what the agent knew at that point) and
→ the harness component most likely at fault.

Harness components:

- prompt: the system prompt text (instructions and policy
→ wording the model reads)
- control_flow: the agent loop code: how the next action is
→ produced, checks or gates applied to tool calls before they
→ run, retries, enforced confirmations
- config: numeric settings such as temperature, token limits or
→ step limits
- output_plumbing: how tool results and tool descriptions are
→ rendered to the model
- context_mgmt: which parts of the conversation history the
→ model sees
- client_tool: extra tools the harness gives the model (e.g. a
→ checker or calculator)
- skill: instruction files loaded for specific situations
- memory: structured state the harness keeps across steps
- subagent: an extra model call, e.g. a second model reviewing
→ an action

Answer with JSON:

```

{"suspects": [{"step": <int>, "why": "<one sentence>",
→ "component": "<component>",
"action": {"message": "<text>" | {"tool_calls": [{"name":
→ "<tool>", "arguments": {...}]}}}]}

```

Largest-gain variant (post hoc). The rerun of
the all-at-once judges and binary search in §5 uses
the same prompts with one sentence of the intro-
duction replaced. Registered:

The DECISIVE MISTAKE is the earliest agent step such that, had
→ the agent acted correctly at that step and continued
→ normally, the task would most likely have succeeded.

Variant:

The DECISIVE MISTAKE is the agent step whose correction would
→ most likely have rescued the task: had the agent acted
→ correctly at that step and continued normally, the task
→ would most likely have succeeded. If several steps qualify,
→ choose the one where a correction is most likely to make
→ the task succeed, which need not be the earliest.

Repair experiment: notes to the proposer. RRSI's proposer prompt is unchanged. In the
no-attribution group the round's analysis report
is replaced by the first note below; the step

τ^2	steps/dec.	utterance	read	write	transfer
Retail	1165/65	54%/28	34%/1	11%/36	1%/0
Airline	1026/37	55%/25	31%/4	12%/8	1%/0
AppWorld	steps/dec.	docs	read	write	submit
AppWorld	1961/55	27%/3	45%/9	14%/29	5%/9

Table 4: Share of all agent steps of each kind / number of decisive steps of that kind. AppWorld "docs" looks up API documentation; 5 AppWorld decisive steps (9% of steps) are other code.

groups get the second note and one digest per failing trace, `{"task_id", "lens": "failure", "decisive_step": k, "step_excerpt": <the step, at most 600 characters>, "blocker": "failure decided at step k"}`.

No analysis report this round. Read the traces yourself
 $\rightarrow$ (list_traces, then read_trace) to find what went wrong; the
 $\rightarrow$ failing traces are listed first.

No analysis report this round. Read the traces yourself
 $\rightarrow$ (list_traces, then read_trace) to find what went wrong; the
 $\rightarrow$ failing traces are listed first. For each failing trace,
 $\rightarrow$ PER-TASK DIGESTS give decisive_step: the step at which an
 $\rightarrow$ automatic failure-attribution method located the failure.
 $\rightarrow$ Read the trace around that step (read_trace with
 $\rightarrow$ from_step/to_step) to see what went wrong there.

C Secondary Metrics, Step Kinds and Search Budgets

Table 5 gives the secondary metrics, Table 4 the kinds of agent steps and decisive steps, and Table 6 counterfactual search by replay budget.

D Post-hoc Variants and Who&When

Table 7 compares the judges' registered prompts with the largest-gain variant (Appendix B), and Table 8 the gold-deviation rule with other methods on τ^2 .

Who&When. To check that our judges are not unusually weak, we ran the all-at-once, step-by-step and binary-search judges on the 184 logs of Who&When with its authors' prompts and the setting that shows the ground-truth answer (Zhang et al., 2025). They identified the responsible agent in 58–67% of logs with all-at-once judging and the responsible step in 3–25% (Table 9), the same pattern the benchmark reports: agents are found far more often than steps.

E Registration and Deviations

The registration (methods, prompts, metrics, the primary comparison and the budget) was committed before any method ran on the main set. Changes

to the attribution study, each logged before the data it affected were analysed:

- **Ground truth.** The registration left the protocol to be chosen on the pilot by test-retest agreement. After the earliest-step protocol agreed on 42% of pilot pairs, protocol 3 and the rescue-gain metrics replaced it (Addendum 1), while methods were running and before any main-set ground truth existed.
- **Retail top-up.** The first loop's harnesses yielded 86 distinct retail failures under the sampling rule; 14 more were drawn with the same rule from two further runs of the loop, as registered. All methods moved by at most 0.02 when they were added.
- **AppWorld.** Registered in Addendum 2 before any AppWorld failure was drawn. Two corrected replays of one step failed on every attempt because the oracle's code passed a literal ellipsis that AppWorld's request log cannot serialise; they count as corrections not applied. That failure's other corrections at the step rescued nothing and its largest gain was 0.375, so they could not have made it rescuable.

Repair experiment. Registered separately, before any attribution was run on its traces and before any candidate was drafted. As checks, the RRSI-analysis group's mean gain (-0.34) matched that of the candidates the original rounds had drafted from the same evidence (-0.42), and re-deploying each incumbent moved its held-out success by -2.5 to $+3.3$ points against its earlier deployment, within the noise of 240 episodes. The first state served as a pilot; it exposed no bug, so its cells were kept, as registered. Two operational problems had no effect on data: a lookup by abbreviated commit hash re-invoked finished deployments, which skipped every episode as already done, and a container restart interrupted the deployments, which resumed with finished episodes kept. One secondary test changed. The registered code tested the round-level contrasts by permuting slots, but a round has one accepted candidate, so each cell's value was copied to its two slots and the permutation split the copies between groups, which is not a valid test. We noticed this after the first run of the analysis, which gave $p = .009$ for first write against no attribution, and replaced the test by an exact sign-flip test over the eight state differences ($p \geq .25$);

Method	Model	Exact [95%]	Within 1	Earliest	Share of max	$R(\hat{k})$, all failures
First write	rule	0.38 [0.27, 0.49]	0.62	0.33	0.52	0.23 [0.17, 0.30]
Last step	rule	0.13 [0.07, 0.20]	0.43	0.08	0.25	0.11 [0.07, 0.16]
All-at-once	Flash	0.28 [0.20, 0.36]	0.52	0.31	0.40	0.19 [0.14, 0.23]
All-at-once	Pro	0.31 [0.22, 0.39]	0.52	0.32	0.39	0.17 [0.13, 0.22]
All-at-once	Pro, thinking	0.32 [0.22, 0.42]	0.60	0.36	0.42	0.18 [0.14, 0.23]
Step-by-step	Pro	0.19 [0.13, 0.26]	0.38	0.22	0.25	0.11 [0.07, 0.15]
Binary search	Pro	0.32 [0.24, 0.40]	0.58	0.34	0.43	0.19 [0.15, 0.24]
Root-cause attributor	Pro	0.28 [0.19, 0.37]	0.49	0.31	0.36	0.16 [0.12, 0.20]
RRSI trace digester	Pro	0.21 [0.14, 0.29]	0.41	0.22	0.29	0.14 [0.10, 0.17]
Counterfactual search@40	Pro, replays	0.26 [0.18, 0.35]	0.44	0.31	0.35	0.15 [0.11, 0.19]

Table 5: Secondary metrics, pooled over the three domains. Exact, within one step, earliest rescuing step and share of the largest gain are over rescuable failures; the last column is over all 300 failures.

Budget	Retail	Airline	AppWorld	Pooled	\$
8	0.22	0.13	0.28	0.22	0.0125
16	0.21	0.23	0.27	0.23	0.0184
24	0.22	0.31	0.27	0.26	0.0212
32	0.22	0.31	0.28	0.26	0.0222
40	0.22	0.31	0.28	0.26	0.0222

Table 6: Counterfactual search: mean $R(\hat{k})$ on rescuable failures by replay budget, read off the same runs, and dollars per failure.

estimates and intervals are unchanged.

Post-hoc analyses. Not registered: the re-scoring under other ground-truth choices, the null-replay rates by step position, the decomposition of the null-free score and the retest agreement among failures both runs found rescuable (Table 2, Figure 3; attrib/robustness.py); the judges asked for the largest gain (Table 7; attrib/rescue_prompt.py); the gold-deviation rule (Table 8; attrib/gold_rule.py); the proposers’ reading of traces in the repair experiment (attrib/phaseb_reading.py); and the no-attribution group’s mean without its worst candidate.

Method	Definition	Retail	Airline	AppWorld	Pooled [95%]	Exact	Position	\$	Difference [95%]
All-at-once, Flash	earliest	0.29	0.41	0.25	0.30 [0.24, 0.37]	0.28	0.64	0.0010	
	largest gain	0.36	0.35	0.26	0.32 [0.25, 0.39]	0.32	0.68	0.0010	+0.02 [−0.04, +0.08]
All-at-once, Pro	earliest	0.30	0.34	0.26	0.30 [0.23, 0.36]	0.31	0.63	0.0031	
	largest gain	0.33	0.47	0.33	0.36 [0.29, 0.44]	0.38	0.65	0.0031	+0.07 [+0.01, +0.12]
Binary search, Pro	earliest	0.33	0.34	0.35	0.34 [0.27, 0.41]	0.32	0.69	0.0064	
	largest gain	0.38	0.42	0.32	0.37 [0.29, 0.44]	0.35	0.72	0.0069	+0.03 [−0.04, +0.10]
First write	rule	0.53	0.27	0.39	0.42 [0.33, 0.51]	0.38	0.78	0	

Table 7: Judges asked for the earliest decisive mistake (registered) or for the step whose correction would most likely rescue the task (largest gain, post hoc), on the 157 rescuable failures: mean $R(\hat{k})$, exact agreement with the decisive step, mean relative position of the named step, dollars per failure (mean over domains), and the paired difference in pooled $R(\hat{k})$ between the two definitions with a cluster-bootstrap interval.

Method	Retail (65)	Airline (37)	τ^2 (102) [95%]	Exact	Gold deviation – method [95%]
Gold deviation (rule)	0.58	0.35	0.50 [0.38, 0.60]	0.49	
First write (rule)	0.53	0.27	0.44 [0.32, 0.55]	0.43	+0.06 [+0.01, +0.12]
All-at-once, Pro	0.30	0.34	0.32 [0.23, 0.40]	0.33	+0.18 [+0.02, +0.33]
Binary search, Pro	0.33	0.34	0.33 [0.24, 0.43]	0.34	+0.16 [+0.02, +0.30]
Counterfactual search@40	0.22	0.31	0.25 [0.17, 0.34]	0.25	+0.24 [+0.09, +0.38]

Table 8: The gold-deviation rule (post hoc): the first state-changing call that matches no unused gold action, else the last step (20% of failures). Mean $R(\hat{k})$ on the 102 rescuable τ^2 failures and paired differences with cluster-bootstrap intervals.

Method	Model	Hand-crafted (58)		Generated (126)	
		agent	step	agent	step
All-at-once	Flash	0.59	0.09	0.58	0.21
All-at-once	Pro	0.62	0.03	0.67	0.18
All-at-once	Pro, thinking	0.62	0.15	0.65	0.25
Step-by-step	Pro	0.24	0.19	0.44	0.32
Binary search	Pro	0.26	0.09	0.38	0.04

Table 9: Who&When, with ground truth shown to the judge as in its authors’ setting: accuracy of the responsible agent and step.